

SHE Culture
ISSN: 2736-0350

Providing insights to innovation in research and reliable industry problem solving

The SHE Culture Journal (ISSN 2736-0350) is the key communication and knowledge exchange tool of the "EMT International Knowledge Exchange Forum for Leadership & Behavioral Safety Dissemination" which serves as a knowledge sharing and EHS dissemination media set up in honour of Professor E. Scott Geller (Distinguished Alumni Professor, Center for Applied Behavior Systems, Virginia Tech, Blacksburg, United States of America (USA), Founder, Actively Caring for People (AC4P) & founding partner, Safety Performance Solutions, SPS, USA

A Publication sponsored by the EMT - CREED, International Projects Team
EMT Centre for Research, Environment & Development, Inc.,
Registered & incorporated in the State of Texas, USA
File Number: 806707886
www.environmenttargets.com
An IFTDO AM630 Member
www.iftdo.org

Advisory & Affiliations

GPFIR & IEHSPST
Global Professors for Innovation in Research & Industry EHS Problem Solving Team

EMT-SCUT
The Chronic Unease & Safety Culture Team

Prof. Patricks-E, Chinemerem
Editorial Board Advisor

El Chumo Inc
Andover Park Drive, Houston, Texas, USA

IFTDO @ AM 630 Member
B7 Kestrel Court, Harbour Road, Portishead, BS20 7AN, UK
S-50, Greater Kailash I, New Delhi 110048, India
www.iftdo.org

Professor E. Scott Geller
Alumni Distinguished Professor
Center for Applied Behavior Systems
Virginia Tech University
Blacksburg, VA, USA.

Professor Ludwig Timothy
Appalachian State University,
United States of America
Safety-Doc.com

Catherine C. Patricks
Website Content Administrator

Technical Review, Editorial Team, Publication & Circulation committee

Erica Judith Antonio, PhD
International Review Consultant

Dr. Neil Fredrick
Coordinating Chair

Innocent Ekanem
Expert in Process Safety & Loss Prevention Engineering

Professor Ugwoha Ejikeme
International Review Consultant, Nigeria
Centre for Occupational Health, Safety & Environment- COHSE
University of Port Harcourt, Nigeria

Correspondence
emtsheculturejournal@gmail.com
environmenttargets@yahoo.com
www.environmenttargets.com
www.iftdo.org

Published by:
EMT Centre for Research, Environment Education & Development, Inc.
in collaboration with
Environment & Millennium Targets Ltd

MEMBER
IFTDO
International Federation of Talent Development Organisations
An IFTDO AM 630 Member Organization

This publication has been edited and reviewed by the SHE Culture Journal (ISSN 2736-0350) editorial and technical review team in consultation with the "EMT Prof. E. Scott Geller International Knowledge Exchange Forum for Safety Leadership & Behavioral Safety Dissemination" ***a knowledge sharing and EHS dissemination media set up in honour of Professor E. Scott Geller (Distinguished Alumni Professor, Department of Psychology, Virginia Tech, Blacksburg, United States of America (USA), Founder & Partner, Safety Performance Solutions, SPS, USA***



Celebrating the academic and consulting giant of Behavioural & Psychological Safety after 56 years on Faculty
(GellerFest, 2025)

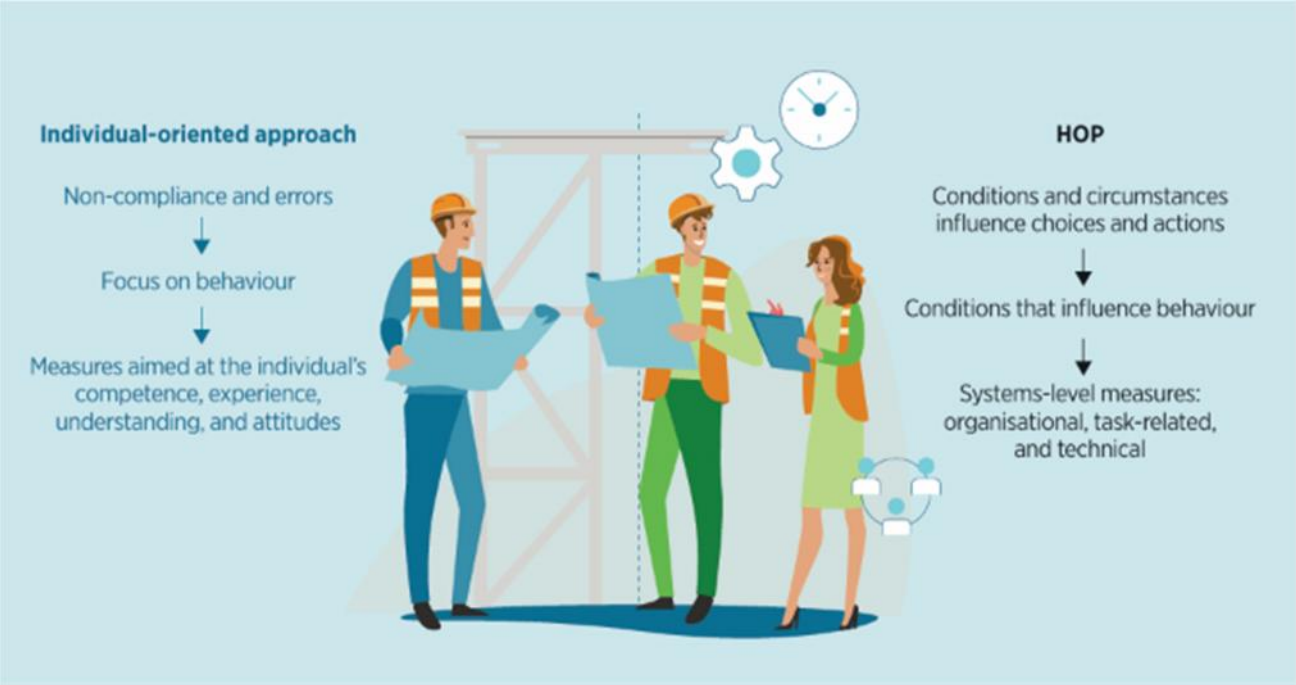
Professor E. Scott Geller (Distinguished Alumni Professor, Department of Psychology, Virginia Tech, Blacksburg, retired in 2025 after 56 years of championing psychological and behavioral safety and contributing to the growth of knowledge and body of literature in this ever-evolving field of learning and academics



GellerFest (May, 2025): Celebrating the distinguished Alumni Professor E. Scott Geller after 56 years at Virginia Tech

A culture of Learning even when nothing goes wrong

Exploring the TEDS Theory



Source: The Federation of Norwegian Industries (2025)

Serious accidents do not happen all the time. Hence, we can say they are rare. However, the conditions that lead to serious incidents are always present in the work place. The difference between days when things go really wrong from days when nothing goes wrong is most often not obvious. The work we do daily is affected by various conditions and circumstances that can make it difficult for workers to comply with requirements and regulations. Hence workers attempt to adjust and workaround to perform the given task. Understanding the contexts and conditions that creates such unexpected and undesired behaviours are essential to make the workplace safe. Tracking daily observations and understanding their trend can help to solve this problem.

Workers make mistakes. Not because they want to make mistakes and harm themselves or others, but because at that point in time, such a choice and actions seem best and offers an opportunity to manage the reality presented by the days task and the conditions around the task. We must assume that most people want to do a good job according to what is expected of them. In practice, this can be challenging. Our decisions and actions are influenced by the systems and conditions around us. To manage challenging conditions, we must understand them first. The analysis of observations and why workers make such deviations and work-arounds are important to keep the work place safe and prevent serious incidents

According to the publication 'Safety, leadership and learning – A practical guide to HOP' by the Federation of Norwegian Industries (2025), Learning from incidents is important. But it is also important to learn from normal work before an incident occurs. This is about learning from what people do as part of their daily work. Most of the jobs we do go well. Here lies a great potential for learning. The conditions that become visible after an incident have often been present in normal work (before the incident) as well. If we become good at learning from normal work, we can identify and manage these conditions and help prevent undesired incidents.

Editorial

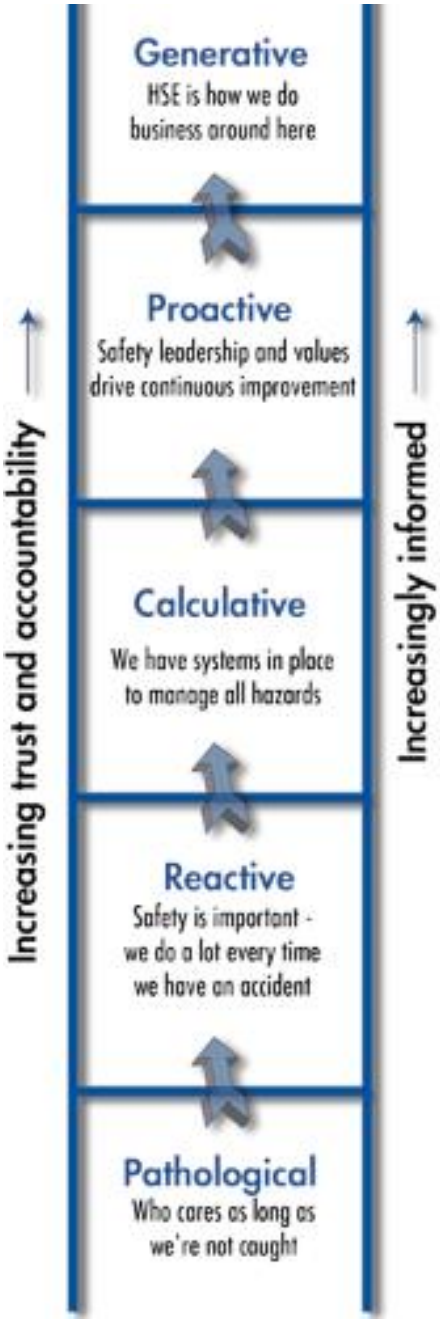
Organizations that have attained the generative stage in their safety culture maturity level understands the value of a learning and reporting culture. When organizations make health and safety a fundamental value and key part of the organizations DNA, the approach to managing workplace risks becomes innovative.

The International Association of Oil and Gas Producers (IOGP), in its report number 435 of year 2010 stated that generative organizations have a high degree of self-sufficiency and strive to understand their entire operating environment. Everyone feels free to highlight both real and potential issues. Workers feel empowered to resolve HSE issues, and leaders provide the support needed.

Learning from everyday normal work is one of the hallmarks of generative organizations. An interesting opportunity to learning is through daily observation of how work is performed and how workers perform their daily task. It is common to observe within pathological and reactive organizations the culture of learning only when things go wrong.

It is essential to query and understand the position of your organization in the HSE Culture ladder. Where do you consider your organization to be in this culture ladder? What key indicators and characteristics informed your decision? What is your organization doing that enabled your classification?

As worksite HSE practitioners, it is essential to take action for each observation made at site. Whether the observation relates to unsafe behaviours, unsafe conditions or safe practices. The TEDS theory which focuses on the Tell, Explain, Describe and Show approach could be beneficial to learning following every workplace observation. Imagine an intervention or engagement with workforce where below questions are asked during an observation. Tell me about the problems with this material, machine, equipment, tool or process. Explain to me what makes your job difficult. Describe the difference between the procedure and the equipment in use. Show me how you do this step of the task.



Source: IOGP (2010), Report No.435

Learn about and use the EMT DSORA App and understand why and how every reported observation helps you to improve your worksite.

ASSESSMENT OF METHODS FOR REPORTING BEHAVIOURAL SAFETY OBSERVATIONS IN SELECTED DOWNSTREAM ENERGY COMPANIES IN NIGER-DELTA.

[1]Onubueze Vincent U., [2]Ugwoha Ejikeme, & [3]Patricks Chinemerem
Centre for Occupational Health, Safety and Environment (COHSE), University of Port Harcourt, Rivers State, Nigeria.

Editor's Note & Commentary: The authors explored methods used in reporting behavioural safety observations and made comparison between the use of an electronic tool and the approach that involves manually writing with paper.

Disclaimer: The data and facts presented here were not subjected to any verification by the SHE Culture journal team and as such are purely research-based findings by the authors within their selected study area.

The Authors:
[1]ONUBUEZE, VINCENT.U., is the lead and corresponding author. He is a PhD student of the Center for Occupational Health, Safety and Environment (COHSE), University of Port Harcourt, Choba, Nigeria.
E-Mail: vincentonubueze@gmail.com, Tel: +234 -80 34721977

[2]UGWOHA. EJIKEME., PhD., is a Professor and the Assistant Director at the Centre for Occupational Health, Safety and Environment (COHSE), University of Port Harcourt, Choba, Nigeria.
E-Mail: ugwohaej@gmail.com , Tel: +234-80 31181050

[3]PATRICKS E.C., PhD., is a professor of Loss Prevention & Human Factors Engineering. He is a member of the academic board and visiting lecturer at the Center for Occupational Health, Safety and Environment, University of Port Harcourt, Choba, Nigeria. He is an industry consultant for safety leadership, human performance, EHS Research & knowledge exchange and a member of EMT/IFTDO International projects committee representing IFTDO in the TC/260, WG-11- ISO 30422:2022 new standard for Learning & Development. He is also a visiting professor to the World Bank Africa Centre of Excellence in Oilfield Chemicals Research (ACE-CEFOR), University of Port Harcourt, Nigeria.
E-Mail: chinemerem.patricks@yahoo.com, Tel: +234- 8038691606

Abstract
This paper assessed the methods for reporting behavioural safety observations in selected downstream oil and gas companies in the Niger-Delta region of Nigeria. Cross-sectional research design was adopted. Three downstream oil and gas companies were sampled using multistage sampling techniques, and sample size of 355 was determined using Fisher's formula. Three operational safety procedures were considered in this study namely, use of personal protection equipment (PPE), equipment handling and general workplace procedures. Survey Questionnaire was designed based on five-point Likert scale for data collection. Data analysis was done using descriptive statistics (Weighted mean score (MS), standard deviation and skewness. The results revealed that: Observational checklist (MS=3.85 > 3.00) and manual reporting technique (MS=3.35 > 3.00)) were the two methods used in reporting behavioural safety observation in the downstream energy sector in Niger-Delta while electronic technique (MS=2.45 < 3.00) are not used for reporting unsafe behaviours in the downstream energy sector in Niger-Delta. The results showed high prevalence of behavioural safety issues observed and reported in the downstream energy companies in Niger-Delta in terms of use of PPE (WMS = 3.64 > 3.00), equipment handling (WMS = 3.08 > 3.00) and general workplace safety procedures (WMS = 3.05 > 3.00). It was concluded that downstream oil and gas firms in Niger-delta uses only traditional technique for reporting unsafe acts and conditions which underscore the need for effective and robust E-reporting techniques in the downstream oil and gas firms in the Niger-Delta.

Key Words; Reporting Methods, Behavioural Safety, Downstream Energy Companies, Niger-Delta

Citation: This paper may be cited as:
Onubueze, V.U., Ugwoha, E., & Patricks, E.C. (2026). Assessment of methods for reporting behavioural safety observations in selected downstream energy companies in Niger-Delta, Nigeria. *SHE Culture Journal*, 8(6), 6-13

INTRODUCTION

Over the years, safety management practices in Nigeria have evolved from a reactive to a proactive approach, with an increasing emphasis on behaviour-based safety (BBS) (Ezekiel-Hart, 2022). This approach focuses on human behaviour change as a key determinant of safety outcomes. BBS is built on the premise that, human factors play a crucial role in accidents. Therefore, the industry has sought to integrate BBS practices to foster a safety culture that goes beyond mere compliance with regulations (Roberts & Geller, 2017; Sharman, 2016). The effectiveness of the BBS initiatives, their alignment with cultural contexts, and the extent of technology integration all warrant careful examination. This research seeks to fill the existing gap in knowledge by carrying out an evaluation on the current status of BBS practices, aiming to identify strengths, weaknesses, and opportunities for improvement.

Behavioural approaches to safety management are now commonplace in most work environments and are designed to improve workplace safety by promoting those behaviours deemed critical to health, safety and risk control (Cooper *et al.*, 2014; Cox & Rycraft, 2016). This is not surprising given that between 80% and 90% of all workplace accidents and incidents can be attributed to unsafe behaviour (Fleming & Lardner, 2015). Fox *et al*. (2017) suggested that effective risk control depends, in part, on the behaviour of individuals at all levels within an organization and has thus highlighted the importance of people focused initiatives. Behaviour-based safety programs are people focused. Most initiatives have a proactive focus, encouraging individuals and their work groups to consider the potential for accident involvement, and their own behaviour as safe versus unsafe before somebody gets hurt (Sulzer, 2018). Colman *et al.* (2017) conducted a review of previous research and reports of interventions, from which it is possible to develop a list of the human factors that influence workplace safety. Many of these factors fall under four overlapping headings: attitudes, perceptions, motivation and behaviour. Jones *et al*. (2008) conducted one of the first studies utilizing behavioural analysis to improve worker safety and concluded that this mode of safety intervention was effective in significantly improving workplace and employee safety performance.

They also reported that not only did employees react favorably towards the behavioural safety programs, but also that the study organization was able to maintain the initiative with a continuing decline in the injury frequency rate. Several safety researchers have supported the findings of Jones, highlighting the effectiveness of applying behavioural analysis techniques to reduce unsafe behaviours and thus lower injury rates (Cooper *et al*., 2014). However, recent evidence appears to suggest that the implementation and sustainability of behavioural safety interventions has been variable and that previously reported improvements in health and safety performance, have lost momentum.

Behavioural safety observation is a proactive approach and can be used to observe and identify people's unsafe acts and conditions. Then, feedback can be provided directly to those who have committed an unsafe act with the aim of modifying their future behavior. It should be acknowledged however that BBS have been widely criticized, there is a proclivity for BBS to neglect the root cause of unsafe behaviour, ignore issues regarding values and attitudes, and can also hide management commitment and inadequacies. To aid this process of reflection and learning, BBS comprises three phases: (1) observation; (2) feedback; and (3) training. Observation is central to monitoring and managing safety behaviour in the workplace. Then, based on individual observable risky behaviour, feedback and training are implemented.

Traditional safety programs put the responsibility of accident prevention and safety coordination on the shoulders of upper management in each company. Upper management generally has to designate a person to act as a safety coordinator and, depending on the size of the company, a group of people to act as the safety team. This person or team has the designation and the responsibility to ensure a safe working environment. An alternative currently being used in industry is behaviour-based safety concept (Geller 2008). BBS involves the practical application of safety procedures based on the real-world behaviours of employees in work situations. Everyone is considered responsible not only for their own safety, but for the safety of others. Unsafe behaviour can trigger accidents and injuries.

Identifying common workplace safety issues enables a company to assess the problem areas and create behavioural-based safety guidelines (Christian *et al.,* 2009; Hermann, *et al.,* 2010; Krause *et al.,* 2009). Behavioural change is not brought about by changing the person, but by changing their perception of accidents and by changing the environment in which they work. It involves observing behaviour and detecting "at-risk" or unsafe activities, followed by directing or modifying behaviour to achieve safe operation (Wirth & Sigurdsson, 2008). Primary techniques in BBS include peer observation and feedback, training and education sessions, behaviour-based incentives, prompts, and goal setting. Thus, this study was carried out to assess the methods of reporting behavioural safety observations in selected downstream energy companies in Niger-Delta.

This study was conducted in the Niger Delta region of Nigeria. Currently, the Niger Delta is mostly recognized as a region that supports extensive oil exploration and production. The Niger Delta basin is widely recognized as the primary pillar of the Nigerian economy due to its substantial and abundant oil reserves. The region also possesses substantial and commercially exploitable resources of natural gas reserves. Petroleum and its byproducts are the primary drivers of the Nigerian economy, accounting for over 90 percent of exports, over 80 percent of the government's annual revenue, and 70 percent of budgeted expenditure. The Nigerian government receives approximately US$ 20 million per day from its crude oil reserves. Currently, Nigeria possesses more than 21 billion barrels of verified oil reserves. Nigeria is the biggest oil producer in Africa and ranks as the sixth most significant exporter of crude oil globally, with majority of its exports being sent to the United States. (Bovwe et al., 2016).

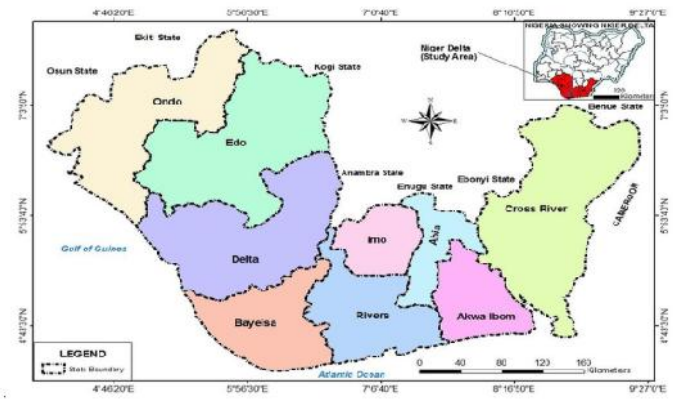


Figure 1: Map of Niger-Delta Showing the study areas

METHODOLOGY

This study adopted mixed-research design which involve combination of cross-sectional design and analytical design (Morse, 2016). Employing both designs would facilitate a thorough comprehension of opinion of the workers and management of companies in the downstream oil and gas sector in Niger-Delta regarding the unsafe acts and conditions that causes accidents in their workplace and also assess the methods for reporting behavioural safety observations.

Population, Sampling technique and sample size determination

The target population of this study was limited to workers in three selected downstream oil and gas enterprises that is located in the Niger Delta region which are actively engaged in storage, processing, distributing or retailing petroleum products. Multi-stage sampling technique and purposive sampling was used to select key respondents from the various companies. The multi-stage or cluster sampling is considered due to the large and dispersed population involved. While purposive sampling was employed to select key respondents from different firms. The respondents were chosen from designated sectors or offices in the downstream companies that have the responsibility of utilizing technology in their operations, alongside other individuals found in the sector.

The sample size was determined using Fishers' sample size determination. This sample size formula is used in a study where the population of the study is unknown and the proportion of the population required for the study can be assumed. The formula is expressed as:

$$n = \frac{Z^2 P}{I^2}$$

Where; n = Sample size, Z = Desired confidence level. In this case, it was taken at 95%. Z value at 95% = 1.96, P = Proportion of the population with desired characteristics. Since it is not known, it was taken as 70%. P value = 0.7, Q = Proportion of the population without the desired characteristics; Q = 1-P, I^2 = Degree of Precision, was taken to be 5% (0.05), $n = 322$.

To cover for non-response, 10% of 322 was used, resulting to 33 extra respondents. Therefore, the minimum estimated sample size for the three sampled companies was 355 at 5% level of confidence. This is in consonant with the rationale that minimum of 10% population (in hundreds) of the study is appropriate for generalization (Elendu, 2010). Three hundred and fifty-five (355) questionnaire was distributed evenly among the three sampled firms X, Y and Z and also equally allotted among HSE officers, field supervisors and field workers as seen in Table 1.

Table 1: Distribution of the study population and sample size

Sn	Company	HSE Officers	Field supervisors	Field workers	Total
1	X	20	20	79	119
2	Y	20	20	78	118
3	Z	20	20	78	118
	Total	**60**	**60**	**235**	**355**

Method and instrument of Data Collection

Survey questionnaire was adopted for collecting data. A fundamental premise of survey quantitative research is the philosophical conviction that our reality is reasonably consistent and uniform, enabling us to measure and analyses it, as well as make extensive speculations about it (Maydeu-Olivares & Millsap 2009).

The survey quantitative approach is focused on providing a numerical representation of response of study respondents suitable for assessing for methods of reporting behavioural safety observations within the downstream petroleum sector in the Niger-Delta Region in areas of Use of Personal Protective Equipment (PPE), equipment handling and general workplace safety procedures.

A self-structured questionnaire was used to obtain data for this study. The questionnaire was explained to the respondents before completion of the questionnaires. The questionnaire was titled "Behavioural Safety Observation Reporting Questionnaire for Downstream Oil sectors" and designed to have five sections; section A and B, Section A was used to obtained information from the respondents on the contemporary methods used in reporting behavioural safety observation in the selected downstream energy companies in Niger-Delta while Section B was used to obtain information on the the prevalent behavioural issues observed and reported in the downstream energy companies in Niger-Delta. The questionnaire is comprised of Conventional 5-point Likert scale comprising of "strongly Agreed (SA) (5), Agreed (A) (4) Undecided (UN) (3), disagreed (D) (2), and strongly disagreed (SD) (1)".

Validity and Reliability of Instrument

The study instrument was validated by experts in Centre for Occupational Health and safety (COHSE), University of Port Harcourt, Rivers State while the reliability was ascertained using Cronbach Alpha test of the pre-test and post-test carried out using 10% of sample size on firms not involved in the survey study but part of the study population. The Cronbach's alpha coefficient of a scale above 0.70 is considered "suitable" for the research as shown in Table 2.

Table 2 Reliability Coefficient of the questionnaire items

	Operation Unit	Number of questionnaire Items	Cronbach alpha coefficient
1	Method of reporting behavioural safety	3	0.73
2	Prevalent behavioural issues observed and reported	35	0.77

Data Analysis

The study employed descriptive statistical methods (means, percentages, weight average). Descriptive statistics was used to fulfil research objectives which involved quantifying the opinion of the respondents on the method of reporting behavioural safety observation in the selected downstream energy companies in Niger-Delta as well as their response on the prevalent behavioural issues observed and reported in the downstream energy companies in Niger-Delta. The analyses were carried out using SPSS version 20.

RESULTS AND DISCUSSIONS

Methods of reporting behavioural safety observation in the selected downstream energy companies in Nigeria

Table 3 shows the results on the methods of reporting behavioural safety observation in the selected downstream energy companies in Niger-Delta. The results revealed that "the used of observational checklist" and "use of manual reporting techniques" are the two-method used in reporting behavioural safety observation in the downstream energy companies in Niger-Delta. Whereas "using electronic reporting tools" is not part of the method used in reporting behavioural safety observation in the downstream energy companies in Niger-Delta. This result suggests that behavioural safety reporting in the energy firm is done manually which underscores the importance of the current study centred on developing electronic tool for reporting behavioural safety concerns in the downstream energy sector.

These finding aligned with the work of Jim et al. (2022) who carried out empirical study to investigate the effectiveness of behavioural-based safety concept in the current day organizations, their study involves 88 international clients who provided over 1.3 million observational data points, including monthly incident and culture survey data.

The outcome of the study revealed that the use of "dedicated observers or manual reporting" is more effective than other procedure for assessment of behavioural-based safety assessment because it encourages employees to participate. They also showed that being observed once a month is more effective than more frequent observations. Further, the results showed how using observers familiar with tasks is more effective than using observers "with a fresh pair of eyes.". They recommended that firms should use BBS only when they have experienced and dedicated people to work in all the stages. The outcome of the study revealed that the use of "dedicated observers or manual reporting" is more effective than other procedure for assessment of behavioural-based safety assessment because it encourages employees to participate. They also showed that being observed once a month is more effective than more frequent observations. Further, the results showed how using observers familiar with tasks is more effective than using observers "with a fresh pair of eyes.". They recommended that firms should use BBS only when they have experienced and dedicated people to work in all the stages.

Table 3. The methods of reporting behavioural safety observation in the selected downstream energy companies in Niger-Delta

S/N	Questionnaire Items	Standard Deviation	Skewness	Weighted Mean Score	Remark
1	The company uses observational checklist to monitor and report unsafe behaviours	0.56	-0.98	3.85	Agreed
2	The company uses electronic tool to monitor and report unsafe behaviours	0.65	0.76	2.45	Disagreed
3	The company uses manual reporting to monitor and report unsafe behaviours	0.86	-0.95	3.35	Agreed
Overall mean score		**0.69**	**-0.39**	**3.21**	**Agreed**

Number of respondents = 332

Prevalent behavioural safety issues observed and reported in the downstream energy companies in Niger-Delta

The prevalent behavioural safety issues observed and reported in the downstream energy companies in Niger-Delta were investigated and the results were presented in Table 4 to Table 6. which covered prevalent behavioural issues observed and reported concerning use of PPE, housekeeping, equipment handling procedures and general workplace safety procedure. Descriptive statistic was used in this study.

On the use of PPE, the results in Table 4 showed that the prevalent behavioural issues observed and reported concerning use of PPE include "Not using Self-Contain Breathing Apparatus (SCBA) when working in confine space Like Inside Fuel Tank", "Not using proper gloves when working on hot or sharp tool and material" "Not using goggles and/or earplug when performing activities that require them like welding of tanks" and "wearing worn-out safety boots when working on slippery floors" while "failing to hook the Full Body Harness when working at height" and "using full body harness with single lanyard when working at height" were not among the prevailing behavioural safety issues observed and reported concerning use of PPE in the downstream energy companies in Niger-Delta. Overall, the result revealed that most of the behavioural safety issues considered in this study are prevailing issues observed and reported in the downstream oil and gas firms in Niger-delta concerning the use of PPE.

On equipment handling procedures, the results in Table 5 showed that "using machinery without protection and/or hand grip" "not giving command or signal when moving working equipment along access path" "not performing pre-inspection on equipment before operation" not supporting equipment properly before working" "using working device beyond their standard usage limit" moving device or equipment unsafely and insecurely" and "modifying working device in a manner that make them unsafe to be used" were among some of the prevailing behavioural safety issues observed and reported concerning equipment handling safety procedure in the downstream energy companies in Niger-Delta while "failing to turn off equipment after they finish operations" and "not applying lock button on the man-lift panel" were not among the prevailing behavioural safety issues observed and reported concerning equipment handling safety procedure in the downstream energy companies in Niger-Delta.

Overall, the result revealed that most of the behavioural safety issues considered in this study are prevailing issues observed and reported in the downstream oil and gas firms in Niger-delta concerning equipment handling safety procedures.

Finally, on the general workplace operations, the results in Table 6 showed that "operating equipment in a fatigue condition" "not giving command or signal when moving working equipment along access path" "entering dangerous areas (dark, slippery, holes and unsecure areas) without permission" "trespassing barricade of the lifting area or heavy equipment working area" "standing within swing perimeter of machine's under operation" "not adopting correct tied when perform lifting" and "performing manual lifting without using the correct procedure" were among the prevailing behavioural safety issues observed and reported concerning general workplace safety procedure in the downstream energy companies in Niger-Delta while "smoking outside of the designated locations and areas" and "keep continuing working under rainy condition" were not among the prevailing behavioural safety issues observed and reported concerning general workplace safety procedure in the downstream energy companies in Niger-Delta. Overall, the results showed most of the behavioural safety issues considered in this study are prevailing issues observed and reported in the downstream oil and gas firms in Niger-delta concerning general workplace safety procedures.

The results aligned with work of Azita et al. (2020) who carried out study aimed at investigating the factors associated with unsafe work behaviour in an Iranian petrochemical company considering the perspectives of workers, supervisors, and safety manager. They conducted qualitative study by recruiting twenty participants from Iranian petrochemical firm using a multi-stage approach, with initial purposive sampling followed by snowball sampling to enhance recruitment. Face-to-face and semi-structured interviews were conducted to gain an in-depth understanding of factors acting as barriers to safe behavior. The results revealed that there were: (i) poor direct safety management and supervision; (ii) unsafe workplace conditions; (iii) workers' perceptions, skills and training; and (iv) broader organizational factors.

The result is contrary to the work of Niciejewska and Obrecht (2020) who carried out empirical study on the impact of behavioural safety on the modification of dangerous behaviours in enterprise in Poland. They conducted survey study among workers and management in the sampled firm using questionnaire. They also prepared and familiarized the employees with observation cards and selected employee group observers. Behavior observation was implemented, and results of the observation were reported. Finally, a second survey was carried out among the employees and management after the entire stage of implementing the BBS project using a questionnaire. And the results revealed that 67% of the surveyed employees indicated that poor organization of work, especially the lack of order at workstations is the main reasons for dangerous behaviours that can lead to accident events. Every fourth employee surveyed also indicates the lack of appropriate breaks at work, which affects the dangerous behavior of employees. The respondents also indicated that the instructions for using machines and devices as insufficiently legible and understandable, which may affect their behavior when performing employee tasks. This proved that the machine and system documentation were not properly prepared. The workers revealed that discussions and conversations during the performance of tasks that require high concentration and lack of reliable preparation for the operation of machines and devices and non-compliance with the instructions are undesirable behavior that negatively affects the safety of work in the enterprise.

Table 4: The prevalent behavioural issues observed and reported concerning use of PPE

S/N	Questionnaire Items	Standard Deviation	Skewness	Weighted Mean Score	Remark
1	Workers fail to hook the Full Body Harness when working at height	0.63	0.82	2.45	Disagreed
2	Workers use Full Body Harness with single lanyard when working at height	0.56	0.78	2.83	Disagreed
3	Workers do not use self-contained breathing apparatus (SCBA) when working in confine space like inside PMS tank.	0.56	-0.93	4.23	Agreed
4	Workers do not use appropriate and suitable hand gloves when working on hot or sharp tool and material	0.64	-0.96	4.35	Agreed
5	Workers do not use goggles and/or earplugs when performing activities that require them like welding of tanks	0.65	-0.93	4.05	Agreed
6	Workers wear worn-out safety boots when working on slippery floors	0.66	-0.89	3.98	Agreed
	Overall Mean Score	0.61	-0.35	3.64	Agreed

Number of respondents = 332

Table 5: The prevalent behavioural issues observed and reported concerning equipment handling procedure

S/N	Questionnaire Items	Standard Deviation	Skewness	Weighted Mean Score	Remark
1	Workers use machinery without protection and/or hand grip	0.93	-0.72	3.15	Agreed
2	Operators do not give any command or signal when moving working equipment along access path.	0.86	-0.76	3.27	Agreed
3	Operators hardly turn off equipment after they finish operations or during equipment maintenance (LOTO).	0.83	0.83	2.43	Disagreed
4	Operators do not perform pre-inspection on equipment before operation.	0.74	-0.92	3.39	Agreed
5	Workers modify working devices in a manner that makes them unsafe to be used	0.83	-0.76	3.15	Agreed
6	Workers do not apply lock button on the man-lift panel	0.78	0.64	2.47	Disagreed
	Overall Mean Score	0.82	-0.44	3.08	Agreed

Number of respondents = 332

Table 6: The prevalent behavioural issues observed and reported concerning general workplace safety procedure

S/N	Questionnaire Items	Standard Deviation	Skewness	Weighted Mean Score	Remark
1	Workers and operators operate equipment in a fatigue condition	0.81	-0.98	4.35	Agreed
2	Operators do not give any command or signal when moving working equipment along access path.	0.86	-0.77	3.28	Agreed
3	Worker smokes outside of the designated locations and areas	0.83	0.91	1.43	Disagreed
4	Workers enter dangerous areas (dark, slippery, confined spaces and unsecure areas) without permission	0.74	-0.92	4.05	Agreed
7	Workers keep continuing working under rainy condition	0.83	0.74	1.33	Disagreed
8	Workers do not adopt standard lifting procedures when performing lifting operations	0.83	-0.76	3.25	Agreed
	Overall Mean Score	0.80	-0.34	3.05	Agreed

Number of respondents = 332

CONCLUSIONS

Following the various outcomes and findings of this study, the following conclusions were drawn: i) the downstream energy companies in Niger-Delta uses only two methods for reporting behavioural safety observation, and the methods are "the use of observational checklist and manual/paper reporting". ii) there are several behavioural safety issues observed and reported in the downstream energy companies in Niger-Delta; concerning use of PPE, the prevalently reported concerns are "Workers not using self-contained breathing apparatus (SCBA) when working in confine space like inside PMS tank. workers not using appropriate and suitable hand gloves when working on hot or sharp tools and materials, workers not using goggles and/or earplugs when performing activities that require them like welding of tanks and Workers wearing worn-out safety boots when working on slippery floors", concerning equipment handling procedures, the prevalently reported concerns are "Workers using machinery without protection and/or hand grip, operators not giving any command or signal when moving working equipment along access path. operators not performing pre-inspection on equipment before operation and workers modify working devices in a manner that makes them unsafe to be used" while concerning general workplace safety procedure, the prevalently reported concerns are "Workers and operators operating equipment in a fatigue condition, operators not giving any command or signal when moving working equipment along access path. Workers entering dangerous areas (dark, slippery, confined spaces and unsecure areas) without permission and Workers not adopting standard lifting procedures when performing lifting operations".

REFERENCES

Bovwe.O., Nwaogazie I L., & Ugbebor, J., (2016) Modeling of Experts Opinions for Efficient Management of Solid Waste in Nigeria, *Asian Journal of Science and Technology* 7 (Issue 05) 2926-2935.

Christian, M.S., Bradley, J.C., Wallace, J.C., & Burke, M. J., (2009). Workplace safety: A meta-analysis of the roles of person and situation factors. *Journal of Applied. Psychology 94*, 1103–1127

Cooper, M. D., Cox, S & Rycraft (2014). Behavioural safety interventions: A review of process design factors. *Professional Safety*, 54(2), 36.

Cox, S & Rycraft, T (2016). Safety, Systems and People, *London, Butterworth Heinemann*.

Elendu, 2010 Elendu, I.C. (2010). Fundamentals of research and statistics for students in human kinetics and other educational disciplines. Port-Harcourt: *Glory of the Latter House Publishers.*

Eziekel-hart, j. C., & Eziekel-hart, Q. J. (2022). Safety and security in selected public sector Organizations in rivers state, Nigeria; the legal perspective. *Safety*, *2*(12).

Fleming, M., & Lardner, R. (2015). Strategies to promote safe behaviour as part of a health and safety management system. BOOK, HSE Books.

Fox, D. K., Hopkins, B. L., & Anger W K. (2017). The Long-Term Effects of a Token Economy on Safety Performance in Open-Pit Mining. *Journal of Applied Behavior Analysis*, Vol. 20.

Geller, E. S. (2008). Understanding behavior-based safety: Step-by-step methods to improve your workplace (Second Edition). Neenah, *WI: J. J. Keller & Associates*.

Hermann, J.A., Ibarra, G.V., & Hopkins, B. L. (2010). A Safety Program That Integrated Behavior-Based Safety and Traditional Safety Methods and Its Effects on Injury Rates of Manufacturing Workers, *Journal of Organizational Behavior Management* 30(1), 6-25.

Jim S., Gennifer L., & Terry M. (2022). Behavior-based safety 2022: today's evidence. *Journal of Organizational Behavior Management* 42(4), 336–359

Jones S., Gennifer L., & Terry M. (2008). Behavior-based safety today's evidence. *Journal of Organizational Behavior Management* 42(4), 336–359

Krause, T.R., Seymour, K.J., &. Sloat K.C.M. (2009) Long-term evaluation of a behavior-based method for improving safety performance: a meta-analysis of 73 interrupted time-series replications, *Safety Science*. 32 1-18

Maydeu-Olivares, A., & Millsap, R. E. (2009). The SAGE handbook of quantitative methods in psychology.

Morse, J. M. (2016). *Mixed method design: Principles and procedures* (Vol. 4). Routledge.

Niciejewska M., & Obrecht M. (2020) Impact of behavioural safety (behavioural-based safety-bbs) on the modification of dangerous behaviours in enterprises. *System Safety Humanity Technology and Facility Environment* 2(1),324-32.

Roberts, S., & Geller, E. S. (2017). Principles of Behavior-Based Safety. *Handbook of safety principles*, 677-710.

Sharman, A. (2016). *From accidents to zero: A practical guide to improving your workplace safety culture*. Routledge.

Sulzer A.B., (2018). The Modification of Occupational Safety Behaviour. *Journal of Occupational Accident*s, 9(3) 23-37.

Wirth, O., & Sigurdsson, O.S., (2008). When workplace safety depends on behavior change: Topics for behavioural safety research. *Journal of Safety Research* 39(6), 589-598.